

INTRAVAGINAL LONGITUDINAL PHYSIOLOGY WEARABLE — SAMPLING, QUALITY INDICES, ARTIFACT REJECTION, AND OUTPUTS (NON-LIMITING)

A. Data Acquisition and Sampling (Non-Limiting)

In various embodiments, an intravaginal longitudinal wearable acquires time-series signals representing tissue-adjacent physiology and internal context over prolonged wear sessions. The device may acquire one or more modalities including bio-impedance (including multi-frequency impedance), capacitive wetness/dielectric sensing, pH sensing (including solid-state and/or ISFET pH), temperature, pelvic activity, distributed pressure/contact sensing, and 6-axis IMU motion/orientation signals. Non-limiting acquisition modes include: 1. continuous sampling during wear; 2. duty-cycled sampling to reduce power consumption while preserving trend fidelity; 3. burst sampling during candidate windows identified by motion stability and contact integrity; and 4. event-triggered sampling responsive to detected transitions, including wetness/dielectric changes, impedance changes, pelvic activity changes, pressure distribution changes, and thermal changes. In certain embodiments, the device time-stamps data on-node and buffers data for later synchronization.

B. Preprocessing and Signal Conditioning (Non-Limiting)

In various embodiments, a computing system performs preprocessing including: 1. filtering and denoising per modality; 2. drift correction and baseline normalization, including temperature compensation and electrode-interface compensation; 3. impedance feature extraction, including magnitude/phase features and frequency-dependent features; 4. wetness/dielectric feature extraction, including trend estimation and discontinuity detection; and 5. pH conditioning and plausibility checks, including stabilization timing and drift modeling in certain embodiments. Processing may be performed on-device, on a user device, and/or in a remote computing environment.

C. Quality Indices and Validity Gating (Non-Limiting)

In various embodiments, the computing system computes one or more quality indices (QI) to accept, down-weight, or exclude windows. Non-limiting QI components include: 1. contact integrity index, derived from distributed pressure stability, coupling continuity, and/or capacitive proximity/contact confirmation where available; 2. motion gating index, derived from IMU magnitude, jerk, and orientation stability; 3. electrode integrity index, derived from impedance plausibility, saturation checks, continuity checks, and/or electrode interface stability; 4. wetness/dielectric plausibility index, derived from smoothness constraints, discontinuity detection, and/or coherence with other modalities; 5. pH sensor quality index, derived from stabilization behavior, drift constraints, and plausibility checks; and 6. thermal plausibility index, derived from temperature stability and coupling dynamics. In certain embodiments, valid windows are defined as those meeting thresholds across at least two indices comprising motion gating and at least one of contact integrity, electrode integrity, wetness plausibility, or thermal plausibility.

D. Artifact Rejection and Robustness (Non-Limiting)

In various embodiments, artifact rejection includes: 1. IMU-informed motion artifact suppression and window rejection; 2. contact artifact detection using pressure distribution discontinuities and coupling instability; 3. electrode artifact detection using impedance discontinuities, saturation, and dropout behavior; 4. moisture artifact detection using abrupt dielectric discontinuities and implausible transitions;

and 5. pH artifact detection using non-physiologic drift rates and stabilization failures. When artifacts are detected, outputs may be reduced-confidence, excluded, or flagged for additional valid windows.

E. Privacy-Preserving Outputs (Node 2) (Non-Limiting)

In various embodiments, Node 2 outputs are presented as abstract indices, zones, distributions, and confidence indicators without explicit anatomical imagery and without explicit anatomical labeling. Non-limiting output families include: 1. hydration / moisture index derived from dielectric and/or capacitive wetness trends; 2. impedance-derived tissue state indices, including frequency-dependent indices and stability indicators; 3. pH trend indices and/or pH variability indices (optionally normalized); 4. pelvic activity indices derived from distributed pressure sensing and/or activity signatures; 5. thermal context tags and temperature-compensated indices; 6. stability/variability indices across validated windows; 7. confidence indicators derived from quality indices and validity gating; and 8. trend curves and distributions representing change-over-time across multi-hour sessions and across repeated sessions.

F. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, Node 2 provides longitudinal outputs for a single user without requiring other nodes. In optional multi-node embodiments, derived outputs may be time-aligned and correlated with other nodes under consent gating and quality thresholds, including correlation with an upper-body context node to condition windows by motion state, thermal state, and wear integrity.